

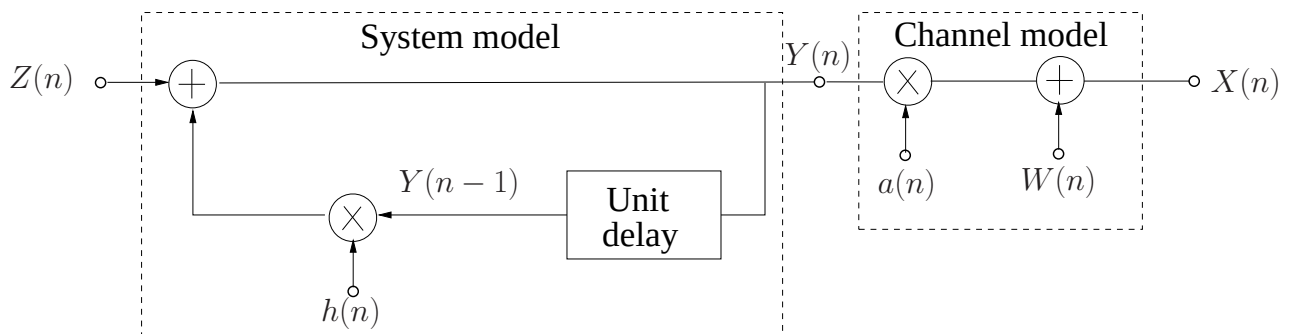
5 Kalman filters

5.1 Scalar Kalman filter

Signal model and Kalman filter to be derived in the sequel:

- **Complete signal model:**

$$\begin{aligned} Y(n) &= h(n)Y(n-1) + Z(n), n = 1, 2, \dots \\ X(n) &= a(n)Y(n) + W(n), n = 1, 2, \dots \end{aligned}$$



- **Scalar Kalman filter**

Prediction step:

$$\hat{Y}(n+1 | n) = h(n+1)\hat{Y}(n | n)$$

$$R(n+1 | n) = h^2(n+1)R(n | n) + \sigma_{ZZ}^2(n+1)$$

$$\hat{X}(n+1 | n) = a(n+1)\hat{Y}(n+1 | n)$$

Updating step:

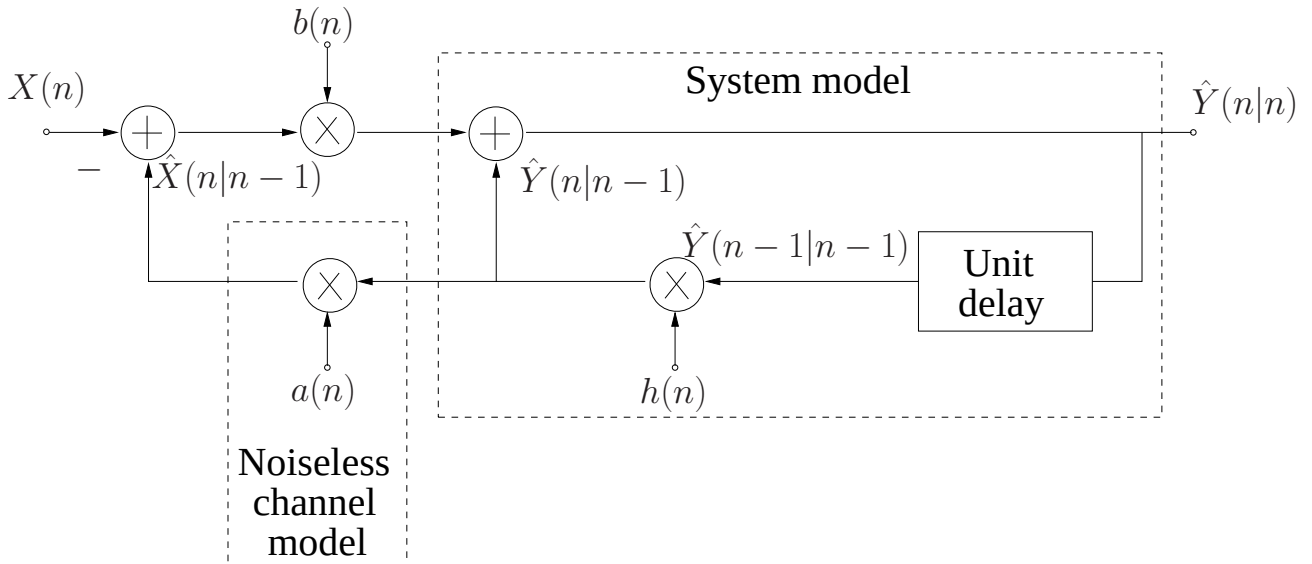
$$\hat{Y}(n+1 | n+1) = \hat{Y}(n+1 | n) + b(n+1)[X(n+1) - \hat{X}(n+1 | n)]$$

$$R(n+1 | n+1) = [1 - b(n+1)a(n+1)]R(n+1 | n)$$

with

$$b(n+1) \equiv \frac{a(n+1)R(n+1 | n)}{a(n+1)^2R(n+1 | n) + \sigma_{WW}^2(n+1)}$$

Block diagram of the scalar Kalman filter:



Initialization:

$$\hat{Y}(0 | 0) = \mu_{Y(0)}$$

$$R(0 | 0) = \sigma_{Y(0)}^2$$

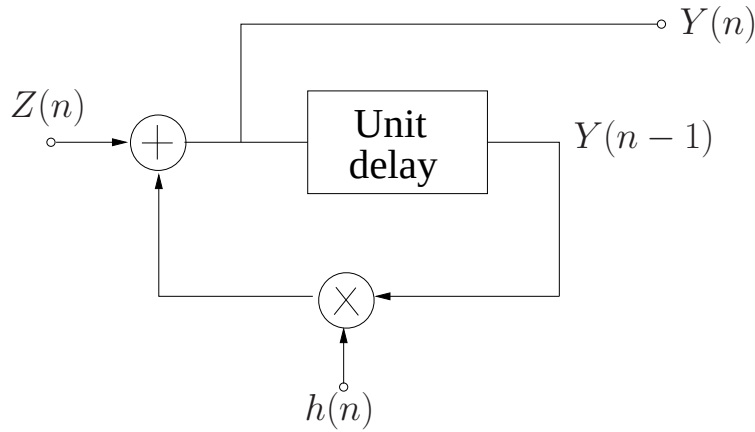
5.1.1 Signal model

- **System model**

$\{Y(n)\}$ is an unobservable sequence which is described by the following *state or system equation*:

$$Y(n) = h(n)Y(n-1) + Z(n), n = 1, 2, \dots \quad (5.1)$$

Block Diagram Representation of (5.1)



Initialization:

$Y(0)$ is a random variable whose expectation $\mu_{Y(0)} \equiv E[Y(0)]$ and variance $\sigma_{Y(0)}^2 \equiv E[(Y(0) - \mu_{Y(0)})^2]$ are known.

Property of the driving process/noise $\{Z(n)\}$:

$\{Z(n)\}$ is an uncorrelated noise with a possibly time-varying variance (non-stationary correlated noise) :

- $E[Z(n)] = 0$
- $E[Z(n)Z(n+k)] = \sigma_{ZZ}^2(n)\delta(k)$

Property of the feedback coefficients $\{h(n)\}$:

$\{h(n)\}$ is a known deterministic sequence.

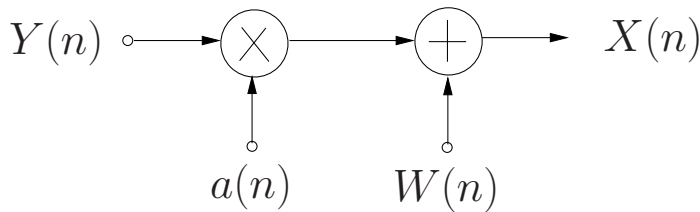
Remark: Provided $\{h(n)\}$ and $\{\sigma_{ZZ}^2(n)\}$ are constant and $\{Z(n)\}$ is a Gaussian random process, then $\{Y(n)\}$ is an AR(1) process.

• **Observation (or channel) model**

The observable sequence $X(n)$ is given by

$$X(n) = a(n)Y(n) + W(n) \quad , n = 1, 2, \dots \quad (5.2)$$

Block diagram representation of (5.2)



Property of the weighting sequence $\{a(n)\}$:

$\{a(n)\}$ is a known deterministic sequence.

Property of the noise $\{W(n)\}$:

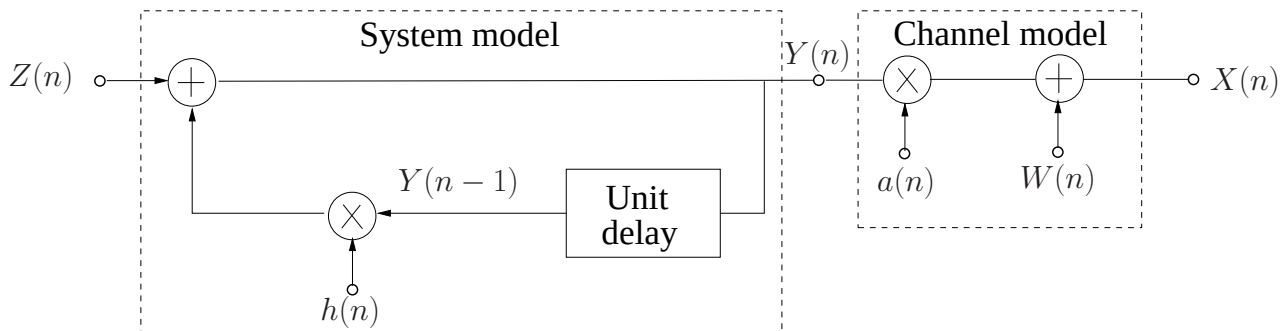
$\{W(n)\}$ is a non-stationary uncorrelated noise:

- $E[W(n)] = 0$
- $E[W(n)W(n+k)] = \sigma_{WW}^2(n)\delta(k)$

• **Additional "weak independence" assumptions**

$Y(0)$, $\{Z(n)\}$, and $\{W(n)\}$ are uncorrelated.

• **Block diagram of the complete signal model**



5.1.2 Recursive implementation of the LMMSEE

- **Objective:**

To find a **recursive implementation**¹ of the LMMSEE of $Y(n)$ based on the observation of $X(1), \dots, X(n)$.

- **Recursive implementation:**

We need the following definitions:

– $\hat{Y}(n | n) \equiv \text{LMMSEE of } Y(n) \text{ based on the observation of } X(1), \dots, X(n)$

Estimation of $Y(n)$.

– $\hat{Y}(n + 1 | n) \equiv \text{LMMSEE of } Y(n + 1) \text{ based on the observation of } X(1), \dots, X(n)$

One-step prediction of $Y(n + 1)$ at time n .

– $\hat{X}(n + 1 | n) \equiv \text{LMMSEE of } X(n + 1) \text{ based on the observation of } X(1), \dots, X(n)$

One-step prediction of $X(n + 1)$.

Recursive implementation of the LMMSEE of $Y(n)$:

$$\underbrace{\hat{Y}(n + 1 | n + 1)}_{\substack{\text{Estimation} \\ \text{at time } n + 1}} \equiv \mathcal{LF}(\underbrace{\hat{Y}(n | n)}_{\substack{\text{Estimation} \\ \text{at time } n}}, \underbrace{X(n + 1)}_{\substack{\text{Observation} \\ \text{at time } n + 1}})$$

where $\mathcal{LF}$ denotes a linear function to be found.

¹See Section 5.3 for an example of a recursive estimator.

- **We shall show:**

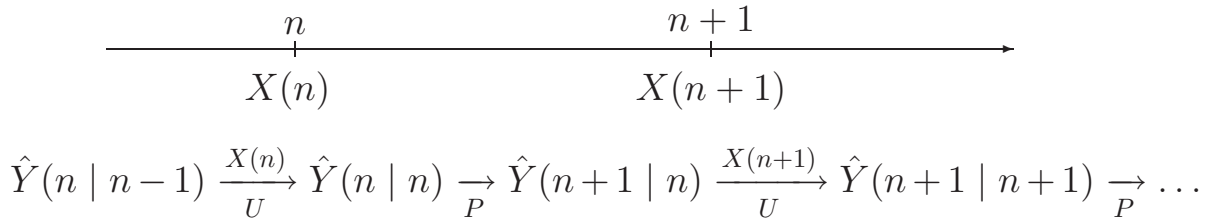
1. Such a recursive implementation of the LMMSEE exists. It is called the ***Kalman Filter***.
2. The recursion is split into two steps:
 - Step 1: ***One-step prediction***:

$$P : \hat{Y}(n | n) \xrightarrow{P} \hat{Y}(n + 1 | n)$$

- Step 2: ***Updating***:

$$U : \hat{Y}(n + 1 | n) \xrightarrow[U]{X(n+1)} \hat{Y}(n + 1 | n + 1)$$

Temporal evolution of the recursive estimation procedure in the Kalman filter:



3. The mean-squared estimation error $E[(Y(n) - \hat{Y}(n | n))^2]$ can also be computed recursively.

We shall need the following definitions:

- $R(n | n) \equiv E[(Y(n) - \hat{Y}(n | n))^2]$
 $\equiv$ mean-squared estimation error at time n
- $R(n + 1 | n) \equiv E[(Y(n + 1) - \hat{Y}(n + 1 | n))^2]$
 $\equiv$ mean-squared one-step prediction error at time n

5.1.3 Derivation of the equations of the Kalman filter

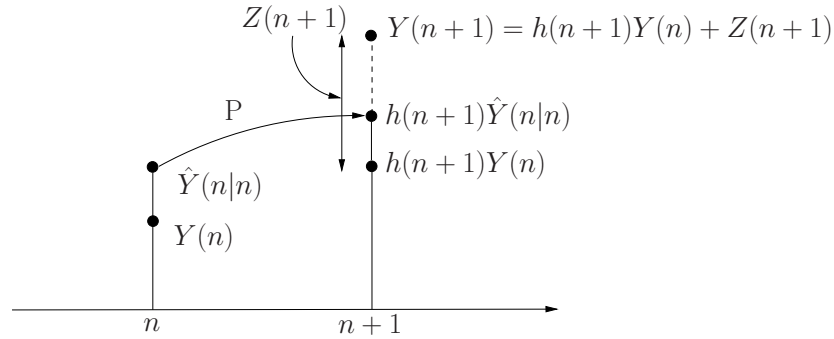
- **Prediction step:**

$$\hat{Y}(n+1 | n) = h(n+1)\hat{Y}(n | n) \quad (5.3)$$

$$R(n+1 | n) = h^2(n+1)R(n | n) + \sigma_{ZZ}^2(n+1) \quad (5.4)$$

Proof of (5.3):

(5.3) follows from the linearity property of the expectation.



Let us show that (5.3) satisfies the orthogonality principle (OP) and therefore is the LMMSEE:

Let $m = 1, \dots, n$:

$$\begin{aligned} & E[(Y(n+1) - \hat{Y}(n+1 | n))X(m)] \\ &= E[\underbrace{(h(n+1)Y(n) + Z(n+1))}_{\text{OP for } \hat{Y}(n|n)} - \underbrace{h(n+1)\hat{Y}(n | n)}_{\text{OP for } \hat{Y}(n|n)}] X(m)] \\ &= h(n+1) \underbrace{E[(Y(n) - \hat{Y}(n | n))X(m)]}_{=0} + \underbrace{E[Z(n+1)X(m)]}_{=0} \\ &= 0 \quad \checkmark \end{aligned}$$

OP for $\hat{Y}(n|n)$

$Z(n+1)$ and $X(n)$
are uncorrelated

Proof of (5.4):

$$\begin{aligned}
 R(n+1 | n) &= E[(Y(n+1) - \hat{Y}(n+1 | n))^2] \\
 &= E[(\overbrace{h(n+1)Y(n) + Z(n+1)} - \overbrace{h(n+1)\hat{Y}(n | n)})^2] \\
 &= E[(\underbrace{h(n+1)[Y(n) - \hat{Y}(n | n)]} + \underbrace{Z(n+1)})^2]
 \end{aligned}$$

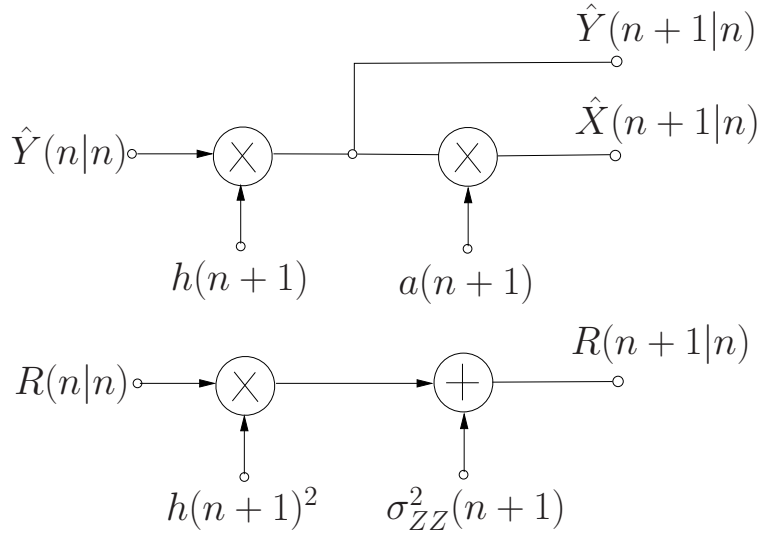
These two random variables are uncorrelated

$$\begin{aligned}
 &= h(n+1)^2 E[(Y(n) - \hat{Y}(n | n))^2] + E[Z(n+1)^2] \\
 &= h(n+1)^2 R(n | n) + \sigma_{ZZ}^2(n+1)
 \end{aligned}$$

With the same argument as that used for the proof of (5.3) we show that

$$\hat{X}(n+1 | n) = a(n+1)\hat{Y}(n+1 | n)$$

Block diagram of the prediction step:



- **Updating step:**

$$\hat{Y}(n+1 | n+1) = \hat{Y}(n+1 | n) + b(n+1)[X(n+1) - \hat{X}(n+1 | n)] \quad (5.5)$$

$$R(n+1 | n+1) = [1 - b(n+1)a(n+1)]R(n+1 | n) \quad (5.6)$$

with

$$b(n+1) \equiv \frac{a(n+1)R(n+1 | n)}{a(n+1)^2R(n+1 | n) + \sigma_{WW}^2(n+1)}$$

Interpretation of (5.5):

$$\hat{Y}(n+1 | n+1) = \underbrace{\hat{Y}(n+1 | n)}_{\substack{\text{One-step} \\ \text{prediction} \\ \text{of } Y(n+1)}} + b(n+1) \underbrace{\left[\underbrace{X(n+1)}_{\substack{\text{New} \\ \text{observation}}} - \underbrace{\hat{X}(n+1 | n)}_{\substack{\text{One-step} \\ \text{prediction} \\ \text{of } X(n+1)}} \right]}_{\substack{\text{Residual error} \\ \text{of } \hat{X}(n+1 | n)}} \underbrace{\quad}_{\text{Correction factor}}$$

Kalman gain:

The coefficient $b(n)$ is called the **Kalman gain** of the filter.

Proof of (5.5) :

We seek an updating equation given by (5.5) and determine $b(n+1)$ so that (5.5) satisfies the orthogonality principle.

1st case: $m = 1, \dots, n$

$$\begin{aligned}
 & E[(Y(n+1) - \hat{Y}(n+1 | n+1))X(m)] \\
 &= \underbrace{E[(Y(n+1) - \hat{Y}(n+1 | n))X(m)]}_{=0 \text{ OP for } \hat{Y}(n+1 | n)} - b(n+1) \underbrace{E[(X(n+1) - \hat{X}(n+1 | n))X(m)]}_{=0 \text{ OP for } \hat{X}(n+1 | n)} \\
 &= 0 \quad \checkmark
 \end{aligned}$$

2nd case: $m = n+1$

$$\begin{aligned}
 & E[(Y(n+1) - \hat{Y}(n+1 | n+1))X(n+1)] \\
 &= E[(Y(n+1) - \hat{Y}(n+1 | n))X(n+1)] \\
 &\quad - b(n+1)E[(X(n+1) - \hat{X}(n+1 | n))X(n+1)]
 \end{aligned}$$

We determine $b(n+1)$ such that the above expression vanishes:

$$b(n+1) = \frac{E[(Y(n+1) - \hat{Y}(n+1 | n))X(n+1)]}{E[(X(n+1) - \hat{X}(n+1 | n))X(n+1)]} = \frac{I}{II}$$

→ **Computation of I:**

$$\begin{aligned}
I &= E[(Y(n+1) - \hat{Y}(n+1 | n)) X(n+1)] \\
&= E[(Y(n+1) - \hat{Y}(n+1 | n)) \overbrace{(a(n+1)Y(n+1) + W(n+1))}] \\
&= a(n+1) \underbrace{E[(Y(n+1) - \hat{Y}(n+1 | n))Y(n+1)]}_{=0} \\
&\quad = E[(Y(n+1) - \hat{Y}(n+1 | n))^2] \\
&\quad \uparrow OP \Rightarrow E[(Y - \hat{Y})Y] = E[(Y - \hat{Y})^2] \\
&\quad + \underbrace{E[(Y(n+1) - \hat{Y}(n+1 | n))W(n+1)]}_{=0} \\
&\quad \quad \quad Y(n+1) - \hat{Y}(n+1 | n) \text{ and } W(n+1) \text{ are uncorrelated} \\
&\quad \quad \quad \text{because } Y(n+1) - \hat{Y}(n+1 | n) \text{ depends on} \\
&\quad \quad \quad Y(0), Y(1), \dots, Y(n), Y(n+1), X(1), \dots, X(n) \\
&= a(n+1)E[(Y(n+1) - \hat{Y}(n+1 | n))^2] \\
I &= a(n+1)R(n+1 | n)
\end{aligned}$$

→ **Computation of II:**

$$\begin{aligned}
II &= E[(X(n+1) - \hat{X}(n+1 | n))X(n+1)] \\
&= E[(X(n+1) - \hat{X}(n+1 | n))^2] \leftarrow OP \\
&= E[(\overbrace{a(n+1)Y(n+1) + W(n+1)} - \overbrace{a(n+1)\hat{Y}(n+1 | n)})^2] \\
&= E[(a(n+1) \underbrace{(Y(n+1) - \hat{Y}(n+1 | n))}_{\text{these random variables are uncorrelated}} + \underbrace{W(n+1)})^2] \\
&= a(n+1)^2 E[(Y(n+1) - \hat{Y}(n+1 | n))^2] + E[W(n+1)^2] \\
II &= a(n+1)^2 R(n+1 | n) + \sigma_{WW}^2(n+1)
\end{aligned}$$

Proof of (5.6):

$$\begin{aligned}
R(n+1 \mid n+1) &= E[(Y(n+1) - \hat{Y}(n+1 \mid n+1))^2] \\
&\stackrel{OP}{=} E[(Y(n+1) - \hat{Y}(n+1 \mid n+1))Y(n+1)] \\
&= \underbrace{E[(Y(n+1) - \hat{Y}(n+1 \mid n))Y(n+1)]}_{\bigcirc} \\
&\quad - b(n+1) \underbrace{E[(X(n+1) - \hat{X}(n+1 \mid n))Y(n+1)]}_{\star}
\end{aligned}$$

Computation of $\bigcirc$:

$$\begin{aligned}
\bigcirc &\stackrel{OP}{=} E[(Y(n+1) - \hat{Y}(n+1 \mid n))^2] \\
&= R(n+1 \mid n)
\end{aligned}$$

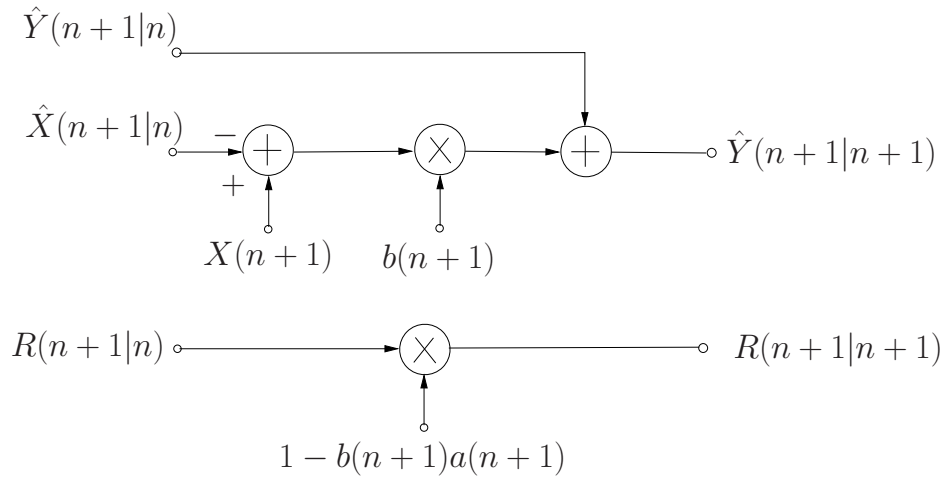
Computation of $\star$:

$$\begin{aligned}
\star &= E\left[\left(a(n+1)Y(n+1) + W(n+1) - a(n+1)\hat{Y}(n+1 \mid n)\right)Y(n+1)\right] \\
&= a(n+1)E[(Y(n+1) - \hat{Y}(n+1 \mid n))Y(n+1)] + \underbrace{E[W(n+1)Y(n+1)]}_{=0} \\
&\stackrel{OP}{=} a(n+1)E[(Y(n+1) - \hat{Y}(n+1 \mid n))^2] \\
&= a(n+1)R(n+1 \mid n)
\end{aligned}$$

Therefore:

$$\begin{aligned}
R(n+1 \mid n+1) &= R(n+1 \mid n) - b(n+1)a(n+1)R(n+1 \mid n) \\
&= [1 - b(n+1)a(n+1)]R(n+1 \mid n)
\end{aligned}$$

Block diagram of the updating step:

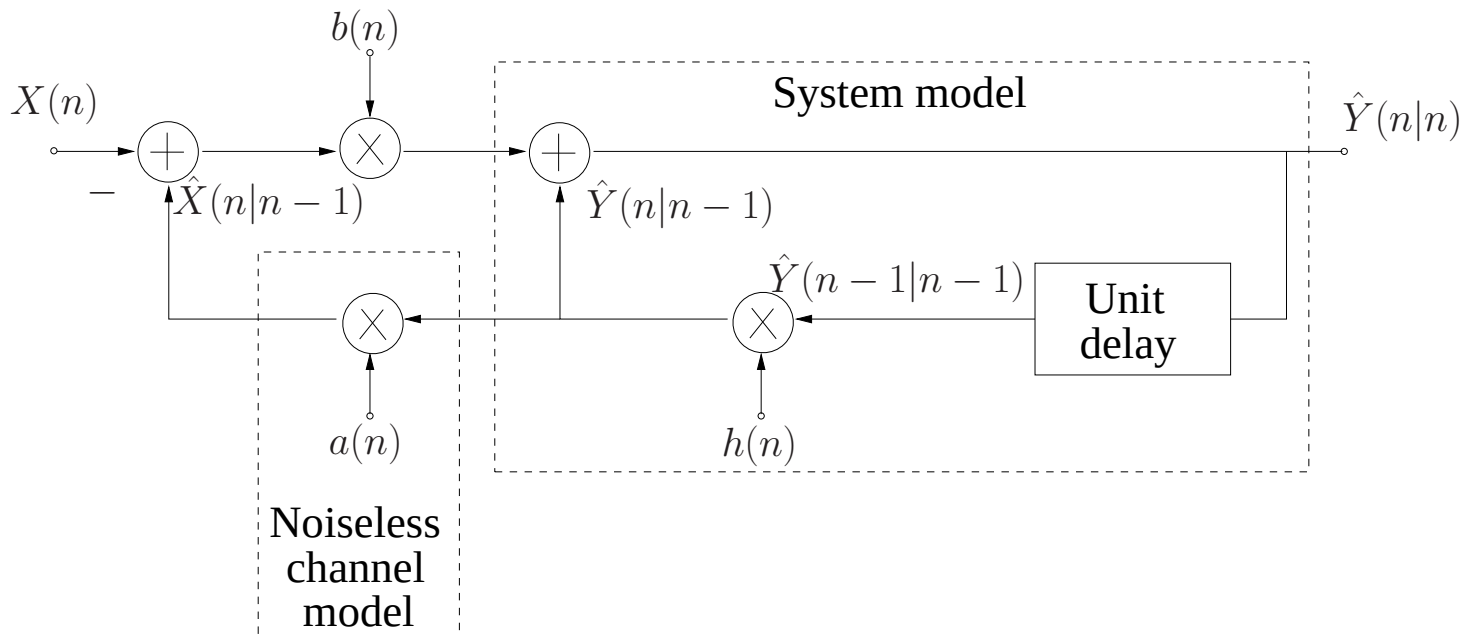


- Initialization:**

$$\hat{Y}(0 | 0) = \mu_{Y(0)}$$

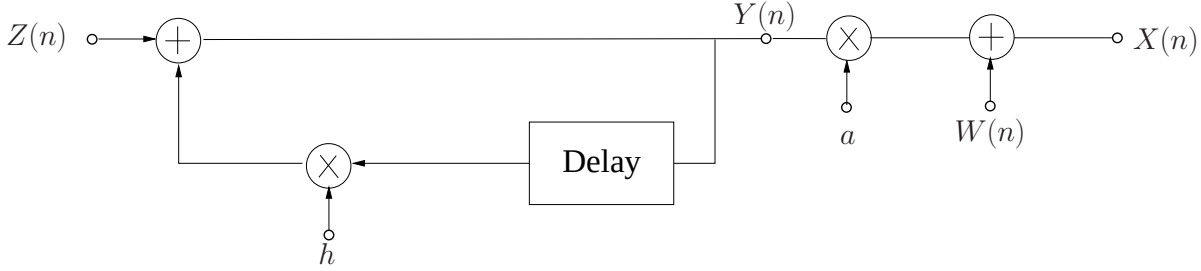
$$R(0 | 0) = \sigma_{Y(0)}^2$$

Block diagram of the Scalar Kalman filter:



5.1.4 Steady-state Kalman filter when the system and channel models are time-invariant

We consider the time-invariant system and channel models as depicted below:



The system-driving process $Z(n)$ and the channel noise $W(n)$ are uncorrelated wide-sense stationary process:

- $E[Z(n)Z(n+k)] = \sigma_{ZZ}^2(n)\delta(k)$
- $E[W(n)W(n+k)] = \sigma_{WW}^2\delta(k)$

Equations of the Kalman filter estimating $Y(n)$:

$$\begin{aligned}
 R(n+1 | n) &= h^2 R(n | n) + \sigma_{ZZ}^2 \\
 b(n+1) &= \frac{a R(n+1 | n)}{a^2 R(n+1 | n) + \sigma_{WW}^2} \\
 R(n+1 | n+1) &= [1 - ab(n+1)] R(n+1 | n)
 \end{aligned}$$

For $n \rightarrow \infty$ the three sequences $\{R(n+1 | n)\}$, $\{b(n)\}$, and $\{R(n+1 | n+1)\}$ converge, i.e.

$$\begin{aligned} R(n+1 | n) &\rightarrow R_{p\infty} \\ b(n) &\rightarrow b_\infty \\ R(n+1 | n+1) &\rightarrow R_\infty \end{aligned} \quad n \rightarrow \infty$$

The Kalman filter converges to its ***steady state***.

The above limits can be calculated by inserting them into the equations of the Kalman filter:

$$R_{p\infty} = h^2 R_\infty + \sigma_{ZZ}^2 \quad (5.7)$$

$$b_\infty = \frac{a R_{p\infty}}{a^2 R_{p\infty} + \sigma_{WW}^2} \quad (5.8)$$

$$R_\infty = [1 - a b_\infty] R_{p\infty} \quad (5.9)$$

Inserting (5.8) into (5.9), we obtain

$$\begin{aligned} R_\infty &= \left[1 - \frac{a^2 R_{p\infty}}{a^2 R_{p\infty} + \sigma_{WW}^2} \right] R_{p\infty} \\ &= \frac{\sigma_{WW}^2 R_{p\infty}}{a^2 R_{p\infty} + \sigma_{WW}^2} \end{aligned}$$

Substituting (5.7) into the last expression yields the so-called steady-state Ricatti equation

$$R_{\infty} = \frac{\sigma_{WW}^2[h^2 R_{\infty} + \sigma_{ZZ}^2]}{a^2[h^2 R_{\infty} + \sigma_{ZZ}^2] + \sigma_{WW}^2}$$

The Ricatti equation is a quadratic equation that can be solved numerically. e.g. by using Newton's method.

Then, $R_{p\infty}$ and b_{∞} follow by inserting the numerical solution for R_{∞} into (5.7) and (5.8), respectively.

Example:

The steady-state solutions for the model with parameter setting

- $h = 0.9$
- $a = 1$
- $\sigma_{ZZ} = \sigma_{WW} = 1$

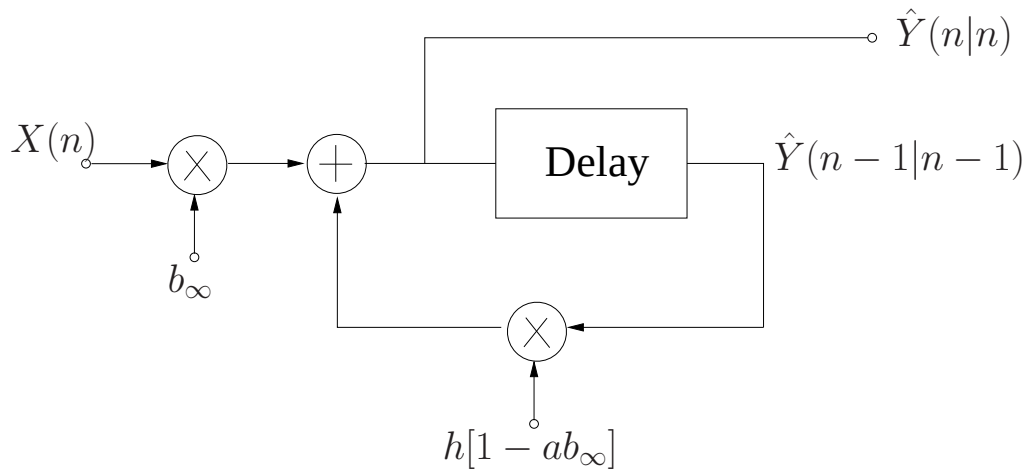
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- $R_{\infty} = 0.5974$
- $R_{p\infty} = 1.4839$
- $b_{\infty} = 0.5974$

Input-output relationship of the steady-state Kalman filter:

$$\begin{aligned}\hat{Y}(n | n) &= h\hat{Y}(n - 1 | n - 1) + b_{\infty}[X(n) - ah\hat{Y}(n - 1 | n - 1)] \\ &= b_{\infty}X(n) + h[1 - ab_{\infty}]\hat{Y}(n - 1 | n - 1)\end{aligned}$$

Block-diagram of the steady-state Kalman filter



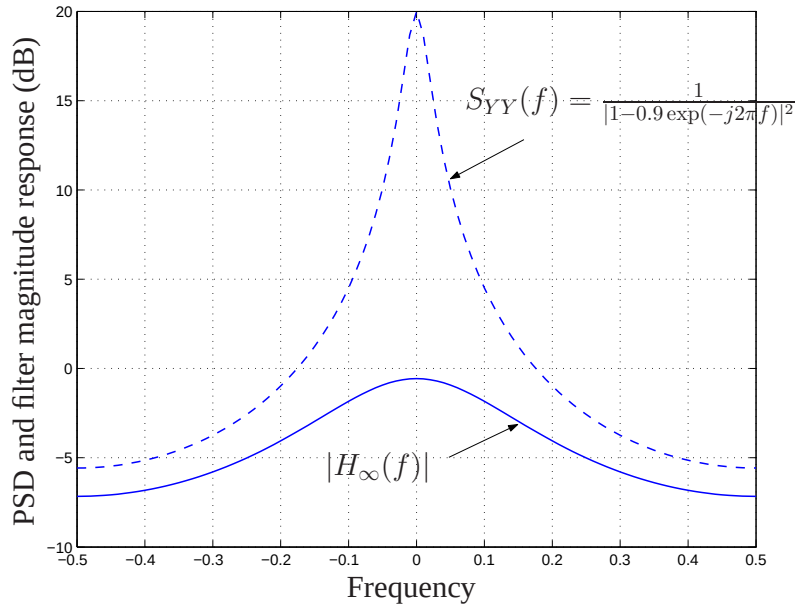
The steady-state Kalman filter is an infinite impulse response (IIR) filter with transfer function

$$H_{\infty}(f) = \frac{b_{\infty}}{1 - h[1 - ab_{\infty}] \exp(-j2\pi f)}$$

$$H_{\infty}(z) = \frac{b_{\infty}}{1 - h[1 - ab_{\infty}]z^{-1}}$$

Example (cont'd):

$$H_{\infty}(f) = \frac{0.5974}{1 - 0.3623 \cdot \exp(-j2\pi f)}$$



Comment:

The steady-state Kalman filter calculates the LMMSEE of $Y(n)$ based on the observation of the sequence $\{X(n)\}$ in the time window $[n, n - 1, n - 2, \dots]$

Hence, the steady-state Kalman filter implements the ***Causal Wiener filter***.

5.2 Vector Kalman Filter

5.2.1 Signal Model

- **System model**

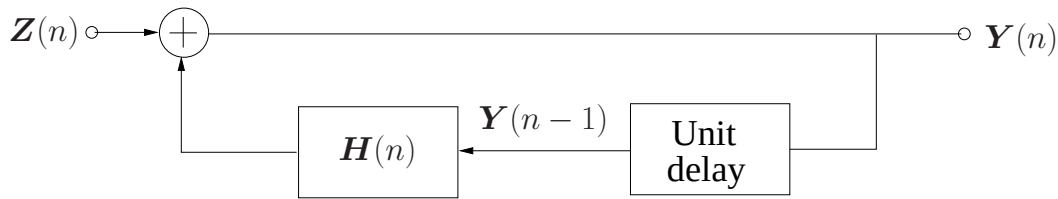
$$\mathbf{Y}(n) = \mathbf{H}(n)\mathbf{Y}(n-1) + \mathbf{Z}(n), \quad n = 1, 2, \dots \quad (5.10)$$

where:

- $\mathbf{Y}(n) = [Y_1(n), \dots, Y_r(n)]^T$: r -dimensional (r -D) random vector.
- $\{\mathbf{Z}(n)\}$: r -D non-stationary uncorrelated noise vector:
 - $E[\mathbf{Z}(n)] = \mathbf{0}$
 - $\sum \mathbf{Z}(n)\mathbf{Z}(n+k) = \mathbf{Q}_Z(n)\delta(k)$
- $\{\mathbf{H}(n)\}$: sequence of known $r \times r$ matrices.

See the example discussed in Section 5.4.

Block diagram:



Initialization

$\mathbf{Y}(0)$ is a random vector specified by its expectation $\mu_{\mathbf{Y}(0)}$ and covariance matrix $\Sigma_{\mathbf{Y}(0)\mathbf{Y}(0)}$.

- **Observation Model**

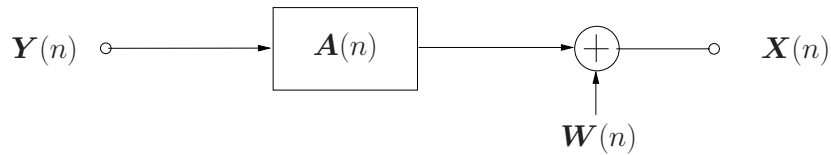
$$\mathbf{X}(n) = \mathbf{A}(n)\mathbf{Y}(n) + \mathbf{W}(n), n = 1, 2, \dots \quad (5.11)$$

where:

- $\mathbf{X}(n) = [X_1(n), \dots, X_s(n)]^T$: s -D random vector.
- $\{\mathbf{W}(n)\}$: s -D non-stationary uncorrelated noise vector with autocovariance

$$\Sigma_{\mathbf{W}(n)\mathbf{W}(n+k)} = \mathbf{Q}\mathbf{W}(n)\delta(k)$$
- $\{\mathbf{A}(n)\}$: sequence of known $s \times r$ matrices.

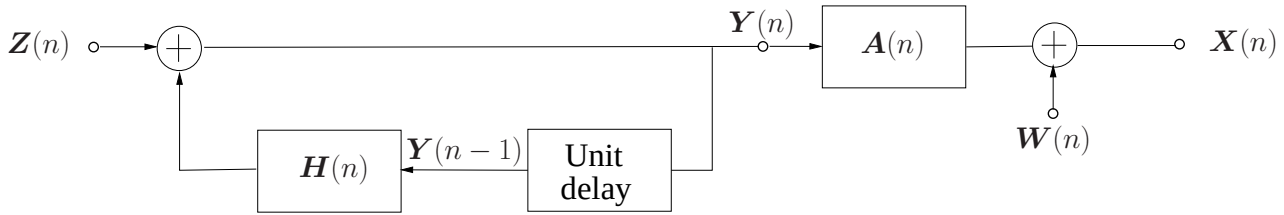
Block Diagram:



- **Additional independence assumption**

$\mathbf{Y}(0)$, $\{\mathbf{Z}(n)\}$, and $\{\mathbf{W}(n)\}$ are uncorrelated.

- **Complete Signal Model**



5.2.2 Equation of the vector Kalman filter

Let us define

- $\hat{\mathbf{Y}}(n | n) \equiv$ LMMSEE of $\mathbf{Y}(n)$ based on $\mathbf{X}(1), \dots, \mathbf{X}(n)$
- $\hat{\mathbf{Y}}(n + 1 | n) \equiv$ LMMSEE of $\mathbf{Y}(n + 1)$ based on $\mathbf{X}(1), \dots, \mathbf{X}(n)$
- $\hat{\mathbf{X}}(n + 1 | n) \equiv$ LMMSEE of $\mathbf{X}(n + 1)$ based on $\mathbf{X}(1), \dots, \mathbf{X}(n)$
- $\mathbf{R}(n | n) \equiv E[(\mathbf{Y}(n) - \hat{\mathbf{Y}}(n | n))(\mathbf{Y}(n) - \hat{\mathbf{Y}}(n | n))^T]$
- $\mathbf{R}(n + 1 | n) \equiv E[(\mathbf{Y}(n + 1) - \hat{\mathbf{Y}}(n + 1 | n))(\mathbf{Y}(n + 1) - \hat{\mathbf{Y}}(n + 1 | n))^T]$

We can apply the same reasoning as used for the scalar Kalman filter to show that the recursive equations of the vector Kalman filter are given as follows.

- **Recursive equations of the Kalman filter**

Prediction Step :

$$\begin{aligned}\hat{\mathbf{Y}}(n+1 | n) &= \mathbf{H}(n+1)\hat{\mathbf{Y}}(n | n) \\ \hat{\mathbf{X}}(n+1 | n) &= \mathbf{A}(n+1)\hat{\mathbf{Y}}(n+1 | n) \\ \mathbf{R}(n+1 | n) &= \mathbf{H}(n+1)\mathbf{R}(n | n)\mathbf{H}(n+1)^T + \mathbf{Q}_Z(n+1)\end{aligned}$$

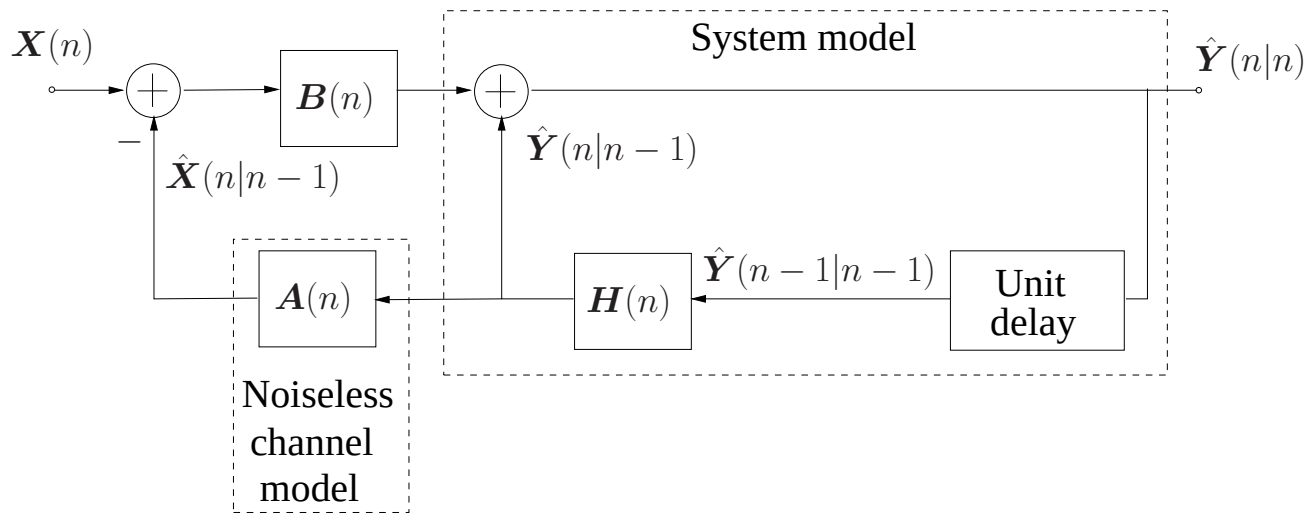
Updating Step :

$$\begin{aligned}\hat{\mathbf{Y}}(n+1 | n+1) &= \hat{\mathbf{Y}}(n+1 | n) + \mathbf{B}(n+1)[\mathbf{X}(n+1) - \hat{\mathbf{X}}(n+1 | n)] \\ \mathbf{R}(n+1 | n+1) &= [\mathbf{I} - \mathbf{B}(n+1)\mathbf{A}(n+1)]\mathbf{R}(n+1 | n) \\ &\text{with the Kalman matrix} \\ \mathbf{B}(n+1) &\equiv \mathbf{R}(n+1 | n)\mathbf{A}(n+1)^T[\mathbf{A}(n+1)\mathbf{R}(n+1 | n)\mathbf{A}(n+1)^T + \mathbf{Q}_W(n+1)]^{-1}\end{aligned}$$

Initialization :

$$\begin{aligned}\hat{\mathbf{Y}}(0 | 0) &= \mu_{\mathbf{Y}}(0) \\ \mathbf{R}(0 | 0) &= \sum_{\mathbf{Y}(0)} \mathbf{Y}(0)\end{aligned}$$

- **Block diagram of the vector Kalman filter**



5.3 Example of a recursive estimator

- **Signal model**

$$X(n) = Y + W(n) \quad n = 1, 2, 3, \dots$$

Where:

- Y is an unknown constant to be estimated based on the observation of $\{X(n)\}$.
- $\{W(n)\}$ is a white noise sequence.

- **Arithmetic mean**

An appealing linear estimator for Y is the ***arithmetic mean***

$$\hat{Y}(n) = \frac{1}{n} \sum_{m=1}^n X(m)$$

Drawback: To compute $\hat{Y}(n)$ based on the above formula, $X(1), \dots, X(n)$ need to be stored. The required memory grows linearly with n .

- **Recursive implementation**

$$\hat{Y}(n+1) = \frac{1}{n+1} \sum_{m=1}^n X(m) + \frac{1}{n+1} X(n+1)$$

$$\hat{Y}(n+1) = \frac{n}{n+1} \hat{Y}(n) + \frac{1}{n+1} X(n+1)$$

This estimator requires storage of one value, i.e. $\hat{Y}(n)$, only.

5.4 Example of a signal model: Target tracking

- **Equations of the movement of a target:**

Position: $U(t) = \int_0^t V(t') dt' + U(0), \quad U(0): \text{initial position}$

Velocity: $V(t) = \int_0^t G(t') dt' + V(0), \quad V(0): \text{initial velocity}$

Acceleration: $G(t)$ is assumed to be white noise.

- **Discrete-time model:**

$$\frac{du}{dt}(t) = V(t) \quad \frac{du}{dt}(nT_s) \approx [U((n+1)T_s) - U(nT_s)]/T_s$$

$$\frac{dv}{dt}(t) = G(t) \quad \frac{dv}{dt}(nT_s) \approx [V((n+1)T_s) - V(nT_s)]/T_s$$

$$U((n+1)T_s) - U(nT_s) = V(nT_s) \cdot T_s \quad T_s: \text{Sampling interval}$$

$$V((n+1)T_s) - V(nT_s) = \tilde{G}(nT_s) \cdot T_s \quad \tilde{G} = \tilde{G}(t) \text{ low-pass filtered with bandwidth } \frac{1}{2T_s}.$$

State model

$$\underbrace{\begin{bmatrix} U(nT_s) \\ V(nT_s) \end{bmatrix}}_{\mathbf{Y}(n)} = \underbrace{\begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}}_{\mathbf{H}(n)} \underbrace{\begin{bmatrix} U((n-1)T_s) \\ V((n-1)T_s) \end{bmatrix}}_{\mathbf{Y}(n-1)} + \underbrace{\begin{bmatrix} 0 \\ T_s \tilde{G}((n-1)T_s) \end{bmatrix}}_{\mathbf{Z}(n)}$$

with

$$\mathbf{Y}(0) = [U(0), V(0)]^T,$$

$$\mathbf{Q}_Z(n) = \begin{bmatrix} 0 & 0 \\ 0 & T_s^2 E[\tilde{G}(nT_s)^2] \end{bmatrix}.$$

Observation model

$$X(n) = U(nT_s) + \underbrace{W(n)}_{\text{Measurement error}}$$

$$X(n) = \underbrace{[1 \quad 0]}_{\mathbf{A}(n)} \mathbf{Y}(n) + W(n)$$

where $W(n)$ is white noise with variance σ_{WW}^2 .